

products by reproducing testimonials as disguised substitute for the test reports issued by competent authorities. Some manufacturers of devices like PowerSave 1200 (PS1200 single-phase), priced at around Rs 12,000, advertise on the Internet providing relevant parameters and certifying agencies and state that their products would qualify for usage in large households with sufficient inductive loads and old equipment (non-energy star rated) and old switchgear/distribution equipment. However, for commercial reasons, they too have placed a number of testimonials from various persons who vouch for the item. But then there is another vociferous community that shows disagreement with its claims.

Words of advice

If the electrical load in your house includes just fans, lights and heaters, in all probability, you do not need to

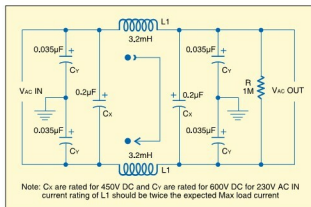


Fig. 4: A typical EMI filter

install a PF-controlling capacitor. As for air-conditioners, pumps, motors and any other high-power inductive loads, these should be individually compensated for power factor degradation (unless already compensated internally) so that only the switched loads are corrected for power factor. To find kVAR values of the PF-correcting

capacitor for individual inductive loads, refer to the 'Power Factor Correction' article in April 2006 issue of EFY.

If you are interested in suppression of transients, surges and lightning effects, read my article on the subject in May and June 2007 issues of EFY. For radio frequency interference/

electromagnetic interference (RFI/EMI) suppression, you may use the filter topology shown in Fig. 4. Capacitors and inductors used in EMI filters typically have the following values:

$$C_x = 0.1 \mu\text{F} \text{ to } 2 \mu\text{F}$$

$$C_y = 2200 \text{ pF to } 33,000 \text{ pF}$$

$$L = 1.8 \text{ mH at } 25\text{A to } 47 \text{ mH at } 0.3\text{A}$$



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Molding



Product Assembly



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